

Design and Optimization of 3D Folded-Core Acoustic Liners for Enhanced Low-Frequency Performance

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Airborne noise with a dominant low-frequency content ($< \sim 500$ Hz) has detrimental effects in many applications, but it is as yet beyond the scope of conventional acoustic noise mitigation techniques using liners, foams, or claddings owing to mass and volume considerations. An alternative approach using liner configurations retaining realistic mass and volume constraints and having innovative 3D folded core geometries is investigated to ascertain its low-frequency noise absorption performance. The relative performance of various candidate 3D folded core designs is compared using a metric termed the *low-frequency performance* (LFP) metric, which is derived from Zwikker-Kosten Transmission Line (ZKTL) theory-based numerical studies. An LFP-based software tool is developed to optimize packing of 3D folded cavities. Experimental verification of absorption coefficient spectra conducted using additively manufactured test articles in normal incidence acoustic impedance tubes yielded good correlation with simulations, verifying the feasibility of this approach. For an optimized design, more than 100 Hz of continuous bandwidth with an absorption coefficient greater than 0.6 is shown to be possible in the 250–400 Hz range with a liner 38.1 mm (1.5 in.) thick. With current additive and hybrid manufacturing techniques attaining critical commercial maturity, 3D folded cavity liners could provide a promising practical solution to mitigate low-frequency airborne noise, especially in aerospace applications.

Nomenclature

C_R	= reflection coefficient
c	= speed of sound in air, m/s
d	= distance between microphones, m
d_{cv}	= diameter of the liner cavity, m
d_{nc}	= diameter of the liner cavity's neck, m
f	= frequency, Hz
g	= distance from liner face to microphone, m
H	= complex acoustic transfer function
h	= height (or length) of a layer of the liner's cavity, m
h_1	= height (or length) of a straight cavity of the liner, m
h_2	= height (or length) of the liner cavity's neck, m
h'_2	= effective height (or length) of the liner cavity's neck, m
k	= free space wavenumber, 1/m
LFP	= low-frequency performance (LFP) metric, Hz/Hz
M	= peak absorption value
N	= number of orifices connected to a single cavity
N_{cv}	= number of cavities in the liner
p_i	= acoustic pressure at i th location
S	= cross-sectional area of the liner cavity's neck, m ²
s	= shear wavenumber, 1/m
T_{ij}	= transmission coefficient
u_i	= particle velocity at i th location
V	= volume of the liner cavity, m ³
α	= absorption coefficient
β	= bandwidth of absorption, Hz

Γ	= propagation constant
γ	= specific heat ratio
ζ	= lower bound of the bandwidth of absorption, Hz
μ	= coefficient of viscosity, Pa · s
ξ_{cv}	= cavity admittance, m ³ /(Pa · s)
ξ_s	= liner surface admittance, m ³ /(Pa · s)
ρ	= density, kg/m ³
ρ_s	= static density of air, kg/m ³
ζ_c	= characteristic impedance, (Pa · s)/m ³
ζ_{cv}	= cavity impedance, (Pa · s)/m ³
ζ_s	= liner surface impedance, (Pa · s)/m ³
σ^2	= Prandtl number
Ω	= surface open area ratio or porosity
ω	= angular frequency, rad/s

I. Introduction

AIRBORNE noise with a dominant low-frequency content can have detrimental effects in many applications. It can excite structural vibration modes, causing increased wear and tear, and even catastrophic failure. It may contribute to cabin noise, which is a source of passenger discomfort in air vehicles; degrade payload integrity; and restrict military mission capabilities when stealth is essential. Moreover, it is a major contributor to environmental noise pollution, which is a major global concern [1–4] especially in commercial aviation. Typically, lower-frequency content ($< \sim 500$ Hz), being less evanescent, persists over longer aerial distances in comparison to higher-frequency content and is as yet beyond the scope of conventional mitigation techniques. Conventional approaches using acoustic liners, foams, or claddings become impractical for low-frequency sound owing to space and weight considerations. One of the most common ways to mitigate noise, especially fan noise through engine ducts [5] or flow-induced noise over airframe structures, is through the use of acoustic liners. Conventional perforate-over-honeycomb liners are tuned to dissipate acoustic energy via viscous mechanisms, create destructive interference for the incident sound wave via reflections within the liner, and sequester tonal energy in the acoustic resonators. The acoustic and structural configurations of liners determine their effectiveness for specific target frequency ranges and application scenarios. The majority of acoustic liners in service in

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aerospace applications do not address these low frequencies mainly because of volume and weight constraints. With the maturation of additive and hybrid manufacturing techniques, there exists an opportunity to explore new approaches to develop lightweight, compact, cost-effective acoustic liners that can effectively mitigate low-frequency noise.

Historical development in the field of acoustic wave propagation was led by pioneers such as Helmholtz [6,7], Kirchhoff [8], Rayleigh [9], Weston [10], Zwikker and Kosten [11], and later by Tijdeman [12]. In the late 1960s, with noise pollution emerging as a significant environmental issue as commercial aviation gained wider prevalence, regulations such as the Noise Control Act of 1972 [1] were implemented. Therefore, improved technologies for airborne noise mitigation began to receive attention from both researchers and practitioners alike. Mangiarotti classified and studied acoustic liners [13,14] as a solution to the problem of reducing noise emission from aircraft engines. Nayfeh et al. [15] investigated the acoustics of aircraft engine duct systems by investigating the dependence of reflection and transmission coefficients on the material properties, duct geometry, and flow-field. While high-frequency components of airborne noise were relatively easy to address using conventional liners, low-frequency components were more challenging to address in practice. An early study by Pratt & Whitney and NASA reported increased effectiveness of acoustic liners for low-frequency applications by coupling locally reacting Helmholtz resonators together with nonlocally reacting concepts [16], although liner thickness was still a barrier to practical application. Baumeister [17] performed a study on optimized multisectioned (or phased) liners and concluded that for low frequencies the segmented liner designs failed to offer a sufficient improvement in performance to justify their replacement of a uniform liner. An early unconventional liner concept called a “silator” that could reduce the overall volume needed by a traditional Helmholtz-style liner was created by evacuating air from an acoustically nonrigid lentiform made from aluminum sheet metal, creating a vacuum chamber in the middle [18]. It was found that the performance of the silator was similar to a conventional liner, while only occupying a hundredth of its volume. Although the silator had good bandwidth, manufacturing complexities for low-frequency designs were a limitation.

Recently, there is renewed interest in airborne noise mitigation technologies due to more stringent noise reduction standards being enacted globally [3,4]. Simultaneously, with the introduction of ultrahigh-bypass-ratio turbofan engines having larger fan diameters in current and next-generation commercial aircraft, blade passage frequencies that are expected to be targeted by acoustic liners would become lower [19]. However, increasing liner thickness is generally not an efficient solution in terms of volume, reaffirming the need for new approaches. Some of these approaches have been for very specific applications. Yu and Chien [20] reported a 10 dB reduction in combustion noise in the frequency range of 400–630 Hz through a folded cavity Helmholtz resonator concept for a hot nozzle center plug. Sugimoto et al. [21] explored a 2D single folded cavity liner to evaluate its absorption characteristics. In comparison to their straight counterparts of the same cavity length, the L-shaped cavities’ behavior was found to be like a deep liner at low frequencies transitioning to that of a shallow liner at higher frequencies. Jones et al. [22] examined three alternate approaches for low-frequency noise absorption. Based on a previous study by Parrott and Jones [23], liner designs consisting of multiple parallel, variable-depth, narrow, and wide chambers including multilayer liners that have mesh caps embedded into its honeycomb core [24] were investigated. Broadband absorption was found to be enhanced using these designs, although obtaining significant absorption below 500 Hz was still a challenge. Use of nontraditional fabrication processes and materials could enable the realization of these variable depth liners.

The emergence of metamaterials concepts as a potential means to attain unusual acoustic properties using engineered structural materials has also resulted in a recent influx of new ideas for low-frequency airborne noise mitigation. Beck et al. [25] studied a concept that seeks to target low-frequency noise by combining a

metamaterials-based design with a traditional acoustic liner design. This design uses large-volume Helmholtz resonators in conjunction with straight cavities to increase low-frequency noise reduction while minimizing liner thickness. The resulting dual-resonant liner has significantly improved performance below 1000 Hz compared with the traditional honeycomb liner, although structural reinforcement may be required for use as primary aerostructures. Fan et al. [26] studied a design featuring stacked membrane-coated perforated plates that showed improved low-frequency sound insulation. Managing manufacturing complexities and ensuring long-term durability of the membranes could make this a promising approach. Ma et al. [27] also showed that membrane-type acoustic metamaterials can be effective to address noise frequencies below 500 Hz. The design features four locally resonant membranes arranged around a circular orifice that acts as a narrow-banded acoustic filter for low-frequency applications in the transmission loss configuration. A modified version could conceivably be adopted for absorptive configurations. Slagle [28] studied a poroelastic, acoustically heterogeneous, and microperforated metamaterial, which showed a significant bandwidth of absorption below 500 Hz. Addressing factors such as overall thickness ($>2''$), use of embedded mass elements, ease of manufacturing, and maintainability could improve its potential for in-service applications. Wu et al. [29] present a low-frequency liner design that uses split tube resonators, resulting in multiple absorption bands at frequencies below 500 Hz. A 3D printed liner design having a single planar coiled chamber was tested by Li and Assouar [30] demonstrating the potential to reach low frequencies with minimal overall thickness. Versaev et al. [19] investigated the performance of spiral cavity liners. The motivation is to shift absorption bandwidths to lower frequencies for a given liner volume. The spiral cavity liner’s insertion loss is comparable to a straight cavity liner having the same cavity depth, which is therefore thicker.

In this study, we demonstrate that it is possible to achieve improved broadband low-frequency ($< \sim 500$ Hz) acoustic absorption performance by efficiently packing tuned combinations of 3D folded cavities within a prescribed volume having practical liner thickness and mass constraints. A software tool is developed to apply Zwikker-Kosten Transmission Line (ZKTL) theory-based simulations to obtain a low-frequency performance metric in order to optimize cavity configurations. Normal incidence tube experiments using 3D printed test articles were conducted to verify the predicted performance. Correlations and deviations between experiments and predictions are discussed with a focus on extension to practical applications.

II. Modeling and Optimization Procedure

The ZKTL model has been discussed in detail by Parrott and Jones [23], and is briefly summarized as follows. One key underlying assumption is that the cavities (or alternatively chambers or channels) with different cavity depths are sufficiently closely packed such that they can be represented with a uniform impedance over their combined surface. This assumption has been shown [31] to hold as long as the spatial separation of the individual cavities is less than one-third of a wavelength at the highest frequency of interest. As shown in the Experimental Methods section, many of the liners considered in this investigation employ folded (bent) cavities. For the purposes of the ZKTL model, these cavities are treated as if they have been “straightened out.” As long as the frequencies of interest are sufficiently low that the sound field is constrained to plane waves, this approach is acceptable.

The ZKTL model for predicting the surface acoustic impedance of an individual liner cavity is briefly described herein with the aid of Fig. 1. This figure provides a sketch of a single cavity of a liner with an air gap (section with height of h_1) and a perforate sheet (thickness of h_2). The thickness of the perforate sheet is exaggerated to demonstrate distinct features of the model used in this study.

The computation is initiated by assuming the backplate to be rigid and impervious. The acoustic pressure and particle velocity are given as

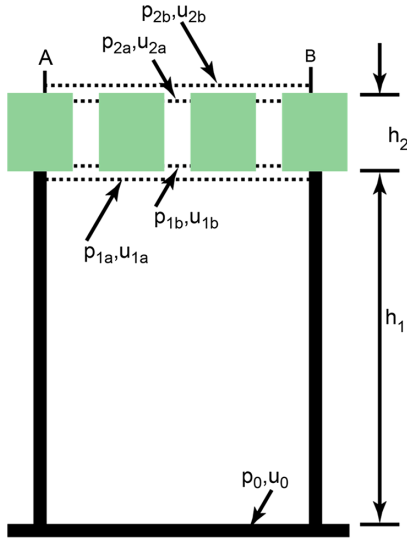


Fig. 1 Sketch of single cavity of a liner with key parameters for ZKTL model.

$$\begin{pmatrix} p_0 \\ u_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1)$$

which yields an infinite impedance at this location. All acoustic pressures and particle velocities are normalized by ρc^2 and c , respectively, where the density of the air and the speed of sound are for the prevailing static pressure and air temperature. Also, impedances are normalized by the characteristic impedance of air, ρc . Changes in the acoustic pressure and particle velocity across the air gap are computed via

$$\begin{pmatrix} p_{1a} \\ u_{1a} \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} p_0 \\ u_0 \end{pmatrix} \quad (2)$$

where the transmission coefficients (T_{11} , T_{12} , T_{21} , and T_{22}) are given by

$$T_{11} = T_{22} = \cosh(k\Gamma h) \quad (3)$$

$$T_{12} = \zeta_c \sinh(k\Gamma h) \quad (4)$$

$$T_{21} = \zeta_c^{-1} \sinh(k\Gamma h) \quad (5)$$

and $k = \omega/c$ is the free-space wavenumber, ω is the angular frequency, and h is the layer thickness. The propagation constant, Γ , and characteristic impedance, ζ_c , are computed as

$$\Gamma = \sqrt{\frac{J_0(i^{3/2}s)}{J_2(i^{3/2}s)}} \sqrt{\frac{\gamma}{n_\Gamma}} \quad (6)$$

$$\zeta_c = \frac{-iJ_0(i^{3/2}s)}{\Gamma J_2(i^{3/2}s)} \quad (7)$$

$$n_\Gamma = \left[1 + \frac{\gamma - 1}{\gamma} \frac{J_2(i^{3/2}\sigma s)}{J_0(i^{3/2}\sigma s)} \right]^{-1} \quad (8)$$

where $i = \sqrt{-1}$ is the imaginary unit, $s = (d_{cv}/2)\sqrt{\rho_s\omega/\mu}$ is the shear wavenumber, and d_{cv} , ρ_s , ω , μ , σ^2 , and γ are the cavity diameter, static density, angular frequency, coefficient of viscosity, Prandtl number, and specific heat ratio, respectively.

The acoustic pressure and acoustic mass flow are assumed constant across the $\{1a, 1b\}$ interface, such that

$$p_{1b} = p_{1a} \quad (9)$$

$$NS_{1b}u_{1b} = S_{1a}u_{1b} \quad (10)$$

where N is the number of orifices (3 in this example) connected to a single air cavity, S_{1a} is the cross-sectional area at the top of the air cavity (denoted by the long dashed line at the top of the cavity), and S_{1b} is the cross-sectional area within a single perforate orifice. Next, the wave propagation within the single orifice is computed using

$$\begin{pmatrix} p_{2a} \\ u_{2a} \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} p_{1b} \\ u_{1b} \end{pmatrix} \quad (11)$$

where the transmission coefficients are given by Eqs. (3–5) and the orifice diameter is used as the “channel diameter.” Again, the acoustic pressure and acoustic mass flow are assumed to be constant across the $\{2a, 2b\}$ interface, such that

$$p_{2b} = p_{2a} \quad (12)$$

$$S_{2b}u_{2b} = NS_{2a}u_{2a} \quad (13)$$

The normalized surface impedance for the cavity is computed as

$$\zeta_{cv} = \frac{p_{2b}}{u_{2b}} \quad (14)$$

Next, the surface impedance spectra of the individual cavities are combined to compute the effective impedance across the liner surface. For this computation, it is preferable to use acoustic admittance ($\xi_{cv} = 1/\zeta_{cv}$). The effective admittance across a selected extent of liner surface is given by

$$\xi_s = \Omega \sum_{i=1}^{N_{cv}} \xi_{cv} \quad (15)$$

where N_{cv} represents the number of cavities that combine to form the liner and Ω is the surface open area ratio (porosity) of the liner. The uniform, effective, surface impedance of the liner is then given by $\zeta_s = 1/\xi_s$.

Simulations using a parallel element liner impedance prediction code developed by Parrott and Jones [23] to predict the acoustic performance for a variety of 2D locally reacting liner configurations form the basis for the development of a 3D folded cavity liner optimization tool. The code is based on the ZKTL theory of sound propagation in channels [11] to predict the surface impedance of a liner as discussed above. This code was later updated and published as a NASA report [32] in 1997. In this approach, individual impedances of an arbitrary set of channels are calculated, and then the area-weighted acoustic admittances of all the channels are summed up in order to find the total admittance of a particular liner design given by Eq. (15). Once the total admittance is known, the total impedance is easily found, and then the absorption coefficient is given by the relationship:

$$\alpha = \frac{4Re(\zeta_s)}{[Re(\zeta_s) + 1]^2 + Im(\zeta_s)^2} \quad (16)$$

where $Re(\zeta_s)$ denotes the real part of the impedance (resistance) and $Im(\zeta_s)$ denotes the imaginary part of the impedance (reactance).

For predicting the overall absorption performance for various straight or folded cavity liner configurations, one only needs to input the total cavity lengths of individual cavities (or channels) within the design into the ZKTL-based code in order to obtain the total admittance and absorption. For the current study, the predicted overall absorption performance for various straight or folded cavity liner configurations was based solely on input of the total cavity lengths of the individual cavities (or channels) into the ZKTL-based code. These calculations did not account for the effects of positioning of individual

cavities with respect to each other. As noted by Schiller and Jones [33], these positioning effects are manifested as changes to the radiation impedance. However, because these radiation impedance corrections have not yet been determined for many of the configurations considered in this study, and because those corrections are generally fairly limited (tend to slightly shift the frequencies of peak absorption), they were not included in the impedance predictions used herein. To optimize a given liner volume to address low-frequency absorption within a specified bandwidth using combinations of 3D folded cavities, effective packaging of these cavities within that volume is essential. An optimization procedure based on the ZKTL-based simulations is employed to achieve this. This procedure is illustrated for a case study for a prescribed low-frequency absorption criterion that also takes into account the constraints imposed by the dimensions and capabilities of the OSU and NASA normal incidence tubes.

The input parameters for the 3D folded cavity optimization case are summarized in Table 1. To set a quantitative criterion to identify a continuous bandwidth, the minimum threshold for significant absorption coefficient, α , is set at 0.6. The targeted center frequency is 350 Hz. The objective then is to maximize the bandwidth of significant absorption ($\alpha > 0.6$) in the vicinity of 350 Hz. The active area of the liner is set to match the internal cross-sectional area of the OSU tube. It is worth noting that the minimum wall and facesheet thicknesses are determined by the minimum thickness needed to ensure acoustic rigidity, which is dependent on the material used for the internal walls. Here, we use a minimum wall thickness of 1.5 mm (0.0591 in.) and facesheet thickness of 1.59 mm (0.0696 in.) determined from experiments on 3D printed ABS structures. The cross-sectional area of each cell is determined by the minimum wall thickness and the number of cells that can be accommodated in each direction for the given external dimensions of the test article. This results in a feasible configuration having a square array of 6×6 cells. Before delving into the details of the optimization, the notations used to identify various features of the design need to be defined. A visual representation of the notations used is shown in the schematic of the top view of the 3D folded liner in Fig. 2. The out-of-plane (or depth) details of the design are described in Sec. III.C. A cell is an individual cavity viewed from the top view (folding is not considered), whereas a cluster is a collection of connected cells that make a single continuous 3D folded cavity. Cells may be full or partial depth. Clusters may contain a partial depth cell. A specific combination of multiple clusters forms a potential configuration for the final test article.

To establish a consistent quantitative measure of low-frequency absorption performance so as to select the optimized configuration, a nondimensionalized metric termed the *low-frequency performance* (LFP) metric was developed. A specific configuration's LFP can be directly calculated from the absorption coefficient spectrum predicted by the ZKTL code using

$$\text{LFP} = \frac{\beta M}{\zeta} \times 100 \quad (17)$$

where β is the lowest continuous frequency bandwidth where the absorption coefficient (α) is greater than the significant absorption threshold (0.6) specified in the input; M is the maximum peak value of the absorption coefficient within the bandwidth, β ; and ζ is the lower bound of the bandwidth, β . A multiplicative factor of 100 is

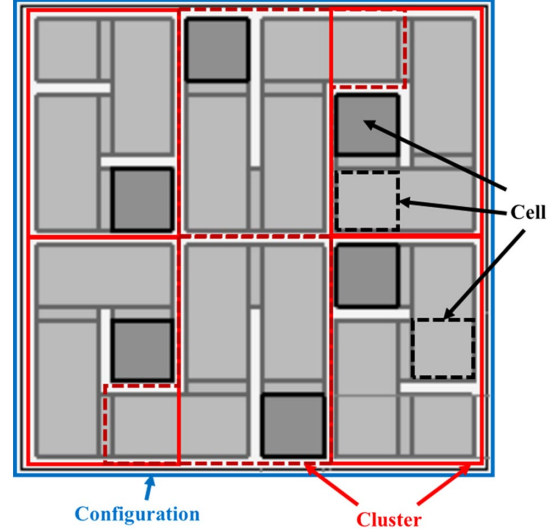


Fig. 2 A 2D visual representation of notations for a 3D folded liner configuration.

also included for presentation purposes. By calculating the LFP from the ZKTL code for each configuration in the iterative optimization process, the configuration with the highest LFP for the targeted frequency range is identified. It is noted that the present choice of parameters for the definition of the LFP is motivated by the liner optimization case study chosen in this study for illustrative purposes. For this case study, maximizing the low-frequency absorption performance in the vicinity of 350 Hz (within the targeted sub-500 Hz range) using a liner 38.1 mm (1.5 in.) thick is considered. In this context, the desirability of attaining absorption above the threshold value (0.6) for as low a frequency as possible while spanning the targeted frequency range makes the choice of β and ζ in addition to M becomes relevant. It is envisioned that, depending on specific design goals, the definition of the LFP could be modified to even include structural or aerodynamic parameters for instance so as to create a metric best suited to the application.

Once the inputs are set, the optimization code is used to determine the possible preliminary configurations having clusters with an integer number of cells that would result in three peaks in the absorption coefficient spectrum. For the constraints of the design for testing, having three distinct peaks in the absorption spectra was identified as producing some of the widest continuous bandwidths. Therefore, in the first step, the total number of cells (36) are apportioned into three distinct cluster types (more than one of each type of cluster can exist in the configuration) and assumed to have an integer number of cells in each cluster.

Thus, the preliminary selection results in three distinct clusters that have three consecutive integer number of cells giving distinct absorption peaks with a continuous bandwidth. For instance, one could apportion a 6×6 configuration into 2 clusters of 5 cells, 2 clusters of 6 cells, and 2 clusters of 7 cells for a total of 36 cells with a configuration notation: $[5(2), 6(2), 7(2)]$. This is the preliminary configuration selected for further optimization from among other 3-cluster configurations based on simulations. These other potential configurations are eliminated based on the input target frequency. Configurations for which the biggest cluster (highest total cavity length) is not sufficient to reach the target frequency are immediately eliminated. Once the viable configuration option with clusters having integral number of cells is known, performance optimization for partial cell lengths resulting from this configuration is then undertaken. For these partial cell depth optimization runs, the total cavity length of the “middle” (6-cell) cluster is fixed, whereas the total cavity lengths for the shorter and longer of the three clusters are differentially varied over one cell depth to identify the final configuration with the best performance. In the present case, the preliminary configuration, $[5(2), 6(2), 7(2)]$, identified above was optimized to the final configuration, $[5^+(2), 6(2), 7^-(2)]$, with total

Table 1 Input parameters for the optimized 3D folded cavity liner design

Parameter	Value	Unit
Volume $[L \times W \times H]$	$50.8 \times 50.8 \times 38.1$ ($2 \times 2 \times 1.5$)	mm^3 (in.)
Targeted center frequency	350	Hz
Significant absorption threshold	0.6	—
Number of cells $[N_l \times N_w]$	6×6	—
Facesheet thickness	1.59 (0.0626)	mm (in.)
Minimum wall thickness	1.5 (0.0591)	mm (in.)

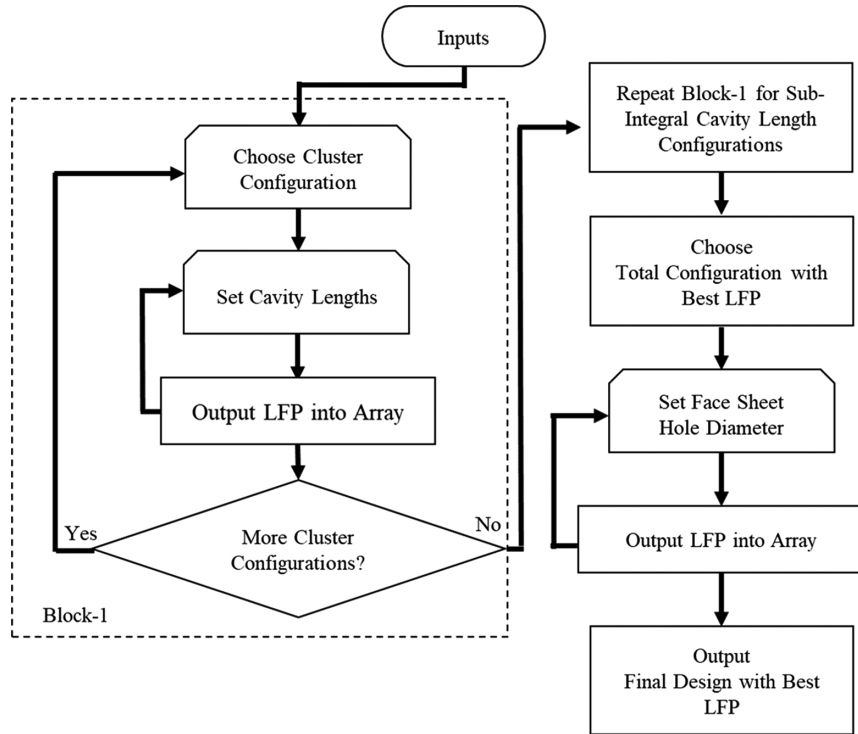


Fig. 3 Flow chart for the 3D folded cavity packing optimization code.

cavity lengths as discussed in Sec. III.C.4. At the same time, the relative partial cell lengths are chosen such that there are no acoustical “dead” volumes in the liner design, which is ideal from a 3D packaging perspective. Further, once the final configuration with optimal cavity lengths is identified, the performance is updated for facesheet porosity for the given facesheet thickness using the LFP to arrive at the final design including the facesheet. The optimal facesheet hole diameter was found to be 2 mm (0.0787 in.) for the configuration considered within the material and fabrication constraints of this study. A facesheet that is 1.59 mm (0.0626 in.) thick with 2-mm-diam holes centered over each cavity opening is used in this design. Figure 3 shows a flow chart depicting the overall optimization process used in this study. Note that the optimization is limited to the design of the core (i.e., the combination of multiple channels with different cavity lengths), whereas the effects of the face sheet are included separately. Clearly, the effects of the face sheet parameters (thickness, hole diameter, porosity) could be included directly into the main optimization (in parallel with selection of the cavity lengths) if the goal were simply to design the optimal absorber. However, if the face sheet thickness and porosity are prescribed through predefined constraints, as was the case for this study, the predominant effect of hole diameter is to adjust the overall resistance. This results in changes to the overall absorption, but has limited effect on the frequency spectrum (i.e., it causes the entire absorption spectrum to shift up or down). The outcome of this optimization process applied to the inputs in Table 1 yielded a configuration for the final design discussed in Sec. III.C.4. As can be seen for the final design, there exists three distinct pairs of clusters, each with a different total cavity length corresponding to three distinct frequency peaks predicted from simulations within the targeted bandwidth. The 3D packing of the clusters is such that there is no dead volume in the liner.

III. Experimental Methods

Two different experimental testing rigs and two variations [34,35] of a standard testing method were employed to evaluate the prototype liner test articles. The first and the most used rig is the normal incidence impedance tube (NIT) at Oklahoma State University. The second is another normal incidence impedance tube at the Liner Technology Facility in NASA Langley Research Center. Both of

these rigs were used to obtain the normal incidence absorption coefficient spectra. All of the designs are tested in the OSU impedance tube, whereas only the Optimized 3D Folded Cavity design is tested at the NASA facility. Experimental NIT results for all designs tested are compared with their respective ZKTL simulation results.

A. Experimental Setup

The normal incidence impedance tube [36] at Oklahoma State University is shown in Fig. 4a. This custom tube was designed to be used for both transmission loss and absorption testing. It consists of an acoustic driver acting as source and controlled using a function generator provided at one end of an acoustic waveguide. The test liner is placed at the other end of the waveguide within an acoustically rigid and sealed sample holder (Fig. 4b). Microphones and a digital acquisition system are used to obtain sound pressure measurements.

For the OSU rig, the waveguide’s length is 914.4 mm (36 in.), with a 63.5 mm × 63.5 mm (2.5 in. × 2.5 in.) internal cross-sectional area with 6.35 mm (0.25 in.) corner fillets. G.R.A.S. microphones with 6.35 mm (0.25 in.) diameter placed at distances of 88.9 mm (3.5 in.) and 139.7 mm (5.5 in.) from the liner face are used to measure the sound pressure. Microphone calibration was performed at the beginning of the experimental process. The normal incidence tube at NASA has a similar configuration (Fig. 5), except that its internal cross section is 50.8 mm × 50.8 mm (2 in. × 2 in.). Tests in the NASA tube are conducted at OASPL of 120 and 140 dB. The NASA tube also uses a rotating two-microphone arrangement for measurements [22]. Thus, test articles of two different dimensions are required for evaluating a liner design in the two tubes, resulting in a difference in the deduced performance. The optimized 3D folded cavity liner test articles are made in the dimensions of both tubes in order to make comparative measurements. Calculations to compute acoustic performance follow the two-microphone method in accordance with either ASTM E1050 [34] or ISO 10534-2:1998 [35]. Noting that the free space wavenumber is given by

$$k = \frac{2\pi f}{c} \quad (18)$$

the reflection coefficient, C_R is calculated as

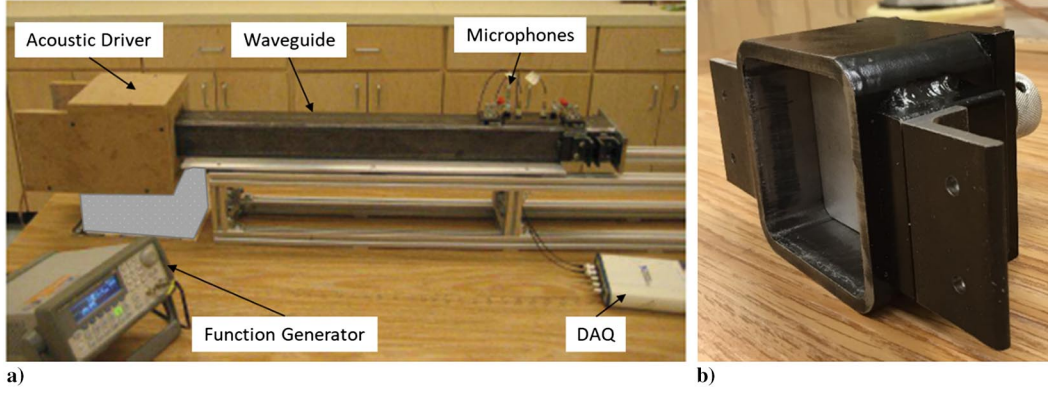


Fig. 4 Photographs of a) the normal incidence impedance tube at OSU and b) its sample holder.

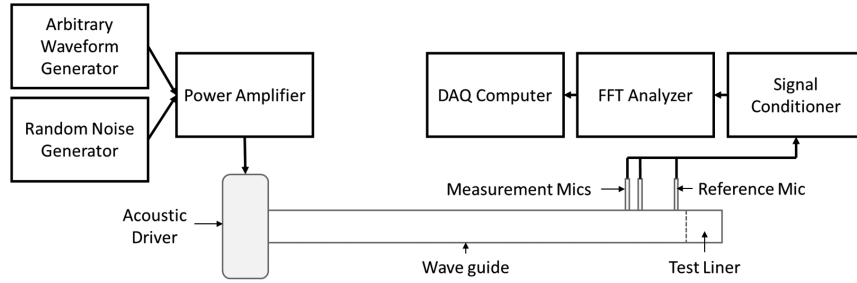


Fig. 5 Schematic configuration of the normal incidence impedance tube at NASA Langley.

$$C_R = \frac{H - e^{-i(d)(k)}}{e^{i(d)(k)} - H} e^{i(2k)(d+g)} \quad (19)$$

where H is the complex acoustic transfer function of the two microphone signals corrected for microphone response mismatch, d is the distance between the two microphones, and g is the distance from the liner face to the closest microphone. Finally, the absorption coefficient α can be calculated as

$$\alpha = 1 - |C_R|^2 \quad (20)$$

Further, the experimental liner surface impedance, ζ_s , can be found using the reflection coefficient as

$$\zeta_s = \rho c \frac{1 + C_R}{1 - C_R} \quad (21)$$

The overall sound pressure level (OASPL) of the broadband white noise signal that was used for the OSU experiments was approximately 110 dB. The frequency range for reliable measurements is ~ 60 to ~ 2600 Hz. The lower frequency limit depends on the spacing of the microphones, and the upper frequency limit depends on the cross-sectional dimension of the waveguide. For each of the test articles tested in the OSU tube, three experimental trials are made, and the results reported are the average of those three trials. Because the NASA Langley Normal Incidence Tube (NIT) has a different internal cross-sectional area, if the same test article is tested in both the OSU and NASA normal incidence tubes, the results will differ in accordance with the change in the ratio of the active area to total area.

B. Test Article Fabrication

All the test articles (except the Helmholtz baseline) are made via additive manufacturing. The Optimized 3D Folded Cavity liner test articles were made in two versions, one in the OSU tube dimensions and another for the NASA tube dimensions. The Helmholtz baseline liner is of sandwich composite construction consisting of a 50.8 mm (2 in.)-thick aluminum honeycomb core bonded to 1.59 mm (0.0626 in.)-thick aluminum facesheets. The honeycomb core has an

average cell diameter of approximately 14.22 mm (0.56 in.). A total of 10 cell cavities were accommodated within the allowable 63.5 mm $\times$ 63.5 mm (2.5 in. $\times$ 2.5 in.) internal cross-sectional area of the OSU tube. Each of the top facesheet holes was drilled into the 1/16''-thick aluminum plate with a 1.59 mm (0.0626 in.)-diam drill bit. The top facesheet was first attached to the core, and then the top facesheet holes were drilled before the bottom facesheet was attached, so as to avoid any debris within the liner cavities.

All additively manufactured test articles were made using fused filament deposition process using desktop 3D printers. Although selective laser sintering process was considered, it was found that the powder could easily get trapped inside any folded cavities during fabrication, which is undesirable. Multiple test articles for the quarter wave baseline liner, U-liner, W-liner, circle spiral liner, square spiral liner, broadband spiral insert liner, and the optimized 3D folded cavity liner and facesheets were 3D printed. ABS plastic was used uniformly as the nominal material for all these test articles. Photographs of selected test articles are shown in Fig. 6.

C. Test Liners

Test liner cases evaluated in this study can be classified into three categories: 1) baseline liner designs, 2) preliminary folded liner designs, and 3) the optimized 3D folded cavity liner design. Details of these designs are discussed in this section.

1. Baseline Liner Designs

To provide a basis for comparison with conventional-core liner designs, two major baseline designs are considered. The first is the straight cavity baseline that has a 5×5 array of straight or full depth cavities within the 3D printed structure as shown in Fig. 7. The design has 25 straight cavities with no folds. Each of these cavities has a depth of 42.97 mm (1.69 in.) and the overall thickness of the liner is 44.45 mm (1.75 in.). The absorption performance of this design is expected to be driven by the quarter wave cancellation phenomenon for this cavity depth. The CAD drawing with dimensions for the straight cavity baseline liner design is shown in Fig. 7.

The second baseline is the Helmholtz baseline liner design. Unlike the 3D printed straight cavity baseline, the Helmholtz baseline is of sandwich composite construction akin to conventional liners. It

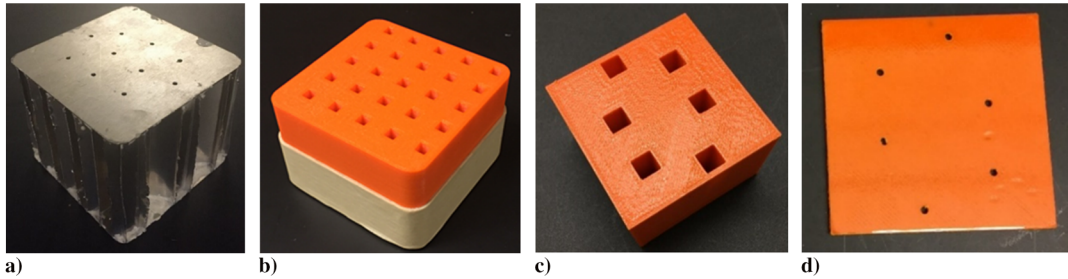


Fig. 6 Photographs of selected test articles: a) Helmholtz baseline, b) W-liner, and c) optimized low-frequency liner and d) its facesheet.

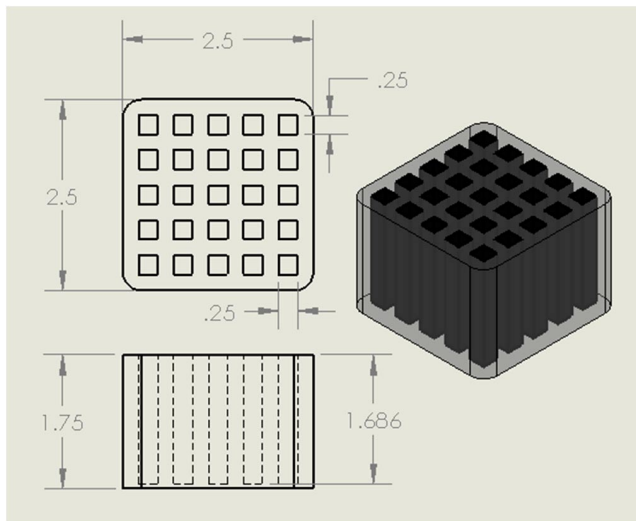


Fig. 7 CAD drawings for straight cavity baseline. All dimensions shown in inches.

consists of a 50.8 mm (2 in.)-thick aluminum honeycomb core provided with a facesheet and back plate as described in Sec. II.B. This baseline provides comparison with performance expected from Helmholtz resonance in a conventional single-degree-of-freedom (SDOF) liner. A photograph of the Helmholtz baseline test article is shown in Fig. 6a. Additional special baseline designs are also considered. These are discussed in ensuing sections.

2. Preliminary Folded Cavity Liner Designs

As a prelude to the optimized 3D folded cavity design, several preliminary folded cavity liner designs were considered. Liner designs with cavities having either right angled (90°) folds (or bends) or full reversal (180°) folds are considered, but packing or bandwidth optimization was not a priority in these designs. First among these is

the U-liner design, which has a set of identical cavities with a single full fold; the objective is to keep the liner volume the same as the straight cavity baseline, but with an increase in each cavity's total length. The design details for the U-liner are shown in Fig. 8a. The U-liner may be termed a *full 2D folded* cavity design given that it has only one full fold per cavity, although one cavity in each row is rotated 90° clockwise as viewed from the liner's top face in order to aid packaging within the tube's dimensions and to retain the symmetric pattern of openings on the liner face. To do this, the cross-sectional area of the postfolded section of this rotated cavity in each row had to be decreased slightly. However, the total depth of each cavity is effectively doubled from the straight cavity baseline design while maintaining the same overall liner volume. The total length of each cavity for the U-liner is 85.60 mm (3.37 in.). In addition to testing the U-liner as a quarter wave resonator without a facesheet, tests were also conducted with a 1.59 mm (0.0626 in.)-thick facesheet with 1.47 mm (0.058 in.)-diam holes added to study the liner's transition to a Helmholtz-type resonator. A special straight cavity baseline liner having the same liner porosity, that is, same number of cavities (25) of the same cavity diameter as the U-liner termed the *U-liner baseline*, was also constructed and tested for comparison.

The third preliminary folded liner design called the "W-liner" extends the U-liner's folded cavity concept to create cavities with three full folds, resulting in an effective cavity length that is about four times the liner's thickness (44.45 mm [1.75 in.]). Two additional folds are introduced to the single full-folded cavity of the U-liner, resulting its cavities having a twisted "W" shape instead of a "U" shape as shown in Fig. 8b. The W-liner design has the same total cavity volume as the straight cavity baseline. The W-liner is expected to deliver better low-frequency absorption due to the quadrupling in the lengths of its cavities in comparison to the original straight cavity baseline design. The total length of each cavity in the W-liner is 166.12 mm (6.54 in.). As before, a special straight cavity baseline liner with the same liner porosity (same number of cavities of the same cavity cross section) as the W-liner, which is termed the *W-liner baseline* is also tested for comparison. Additionally, a 1.59 mm (0.0626 in.)-thick facesheet with 1.47 mm (0.058 in.)-diam holes

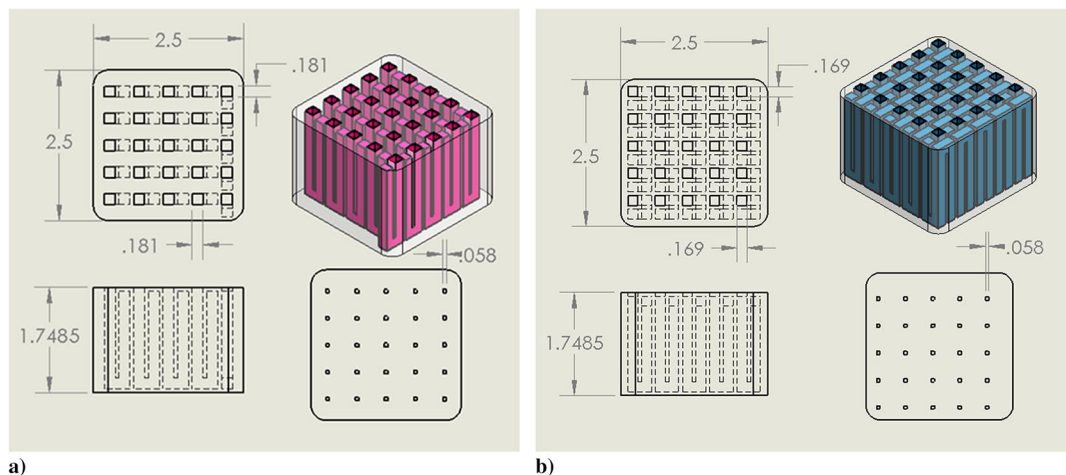


Fig. 8 CAD drawings for a) U-liner and b) W-liner designs. All dimensions shown in inches.

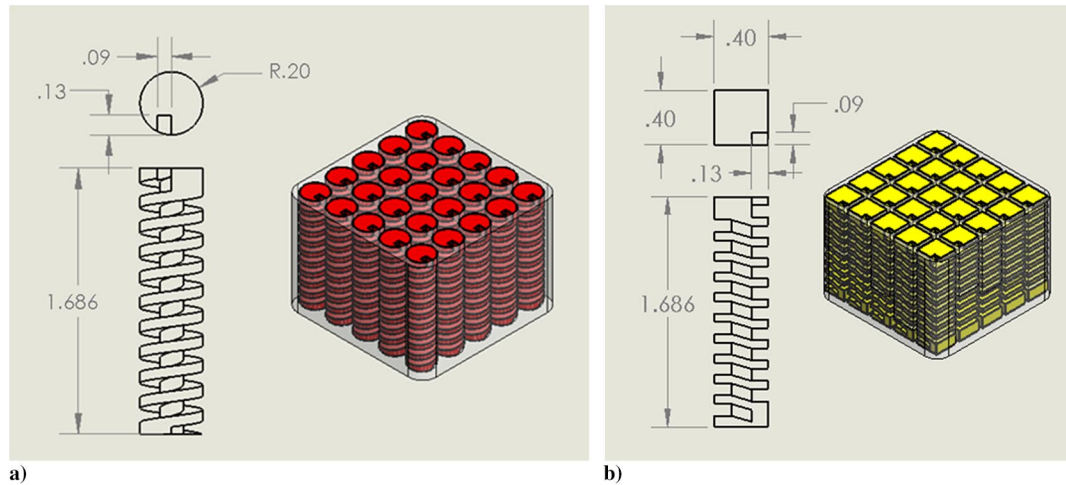


Fig. 9 CAD drawings for a) circle spiral and b) square spiral liner designs. All dimensions shown in inches.

was also added to the W-liner and tested. It is noted that the total liner open area is nearly the same for the U and W liners and their straight cavity baseline liners.

3. Spiral Cavity Liner Designs

To enhance the absorption performance for the lowest frequencies possible, it is desirable to maximize the acoustic path length achievable using the liner cavities. Spiral cavities could provide a means to achieve this. Two types of spiral cavity liner designs are considered from a geometry point of view. The circle spiral design has each cavity path following a spiral path about a cylindrical core. The objective is to create a liner with 25 identical cavities as in the straight cavity baseline, U-liner, and W-liner that has the same total liner volume as these previous liners, but with much higher cavity lengths due to the spiral geometry. By using the spiral geometry, the cavity length of each of the cavities in this design is 33.2% higher than the cavity length of each cavity in the W-liner design. This allows for even lower frequencies to be addressed, with the lowest absorption peak expected to occur around 375 Hz. The circle spiral design and dimensions are shown in Fig. 9a.

An alternative spiral cavity design that is considered is the so-called square spiral design in which the acoustic path is spiraled around a square-shaped core instead of a circular one as shown in Fig. 9a. This design could potentially be easier to manufacture than a circular spiral design using additive manufacturing. Unlike the circle spiral design, the square spiral has right angle turns inside the cavity. The cross-sectional area of the cavity is kept the same as that of the circle spiral design, although the cross section of the cavity in this case is rectangular. The overall cavity length for this design is slightly greater than that for the circle spiral design; therefore, the first peak in

the absorption spectrum is expected to occur around 285 Hz. Square spiral design and dimensions are shown in Fig. 9b.

Taking a cue from the potential of spiral cavities to address lower frequencies, a broadband spiral insert design is considered in order to investigate if it could enhance the low-frequency bandwidth of significant absorption. Five different spiral cavity lengths are arrayed within the liner in this design as shown in Fig. 10. The specific cavity lengths are chosen to distribute their absorption peak locations within a wide bandwidth of interest. However, instead of having circle or square cell geometry, this design uses a hexagonal geometry in order to serve as an insert within a traditional honeycomb core. By using an aluminum honeycomb core as the primary structure into which the 3D printed spiral cavity units are inserted, advantages of having a conventional relatively lightweight yet stiff construction is retained. Because of the hexagonal cell shape, a circular spiral path was chosen to improve volume usage. The open cavity cross-sectional area was kept the same as in the square spiral design, but due to the hexagonal geometry, the maximum cavity length is lower. Further, the overall liner thickness for this insert-type liner is not 44.45 mm (1.75 in.) as for the other 3D printed cases but 50.8 mm (2 in.), due to the thickness of the aluminum honeycomb core used to construct it.

4. Optimized 3D Folded Cavity Liner Design

As a case study, the optimization process for 3D folded cavity liners was implemented to meet a specified low-frequency absorption criterion. The criterion used for this is to maximize the bandwidth of significant absorption ($\alpha > 0.6$) in the vicinity of 350 Hz using combinations of 3D folded cavities of various lengths within a liner of a specified thickness (38.1 mm [1.5 in.]). A conventional straight cavity liner of this thickness would display an absorption peak in the

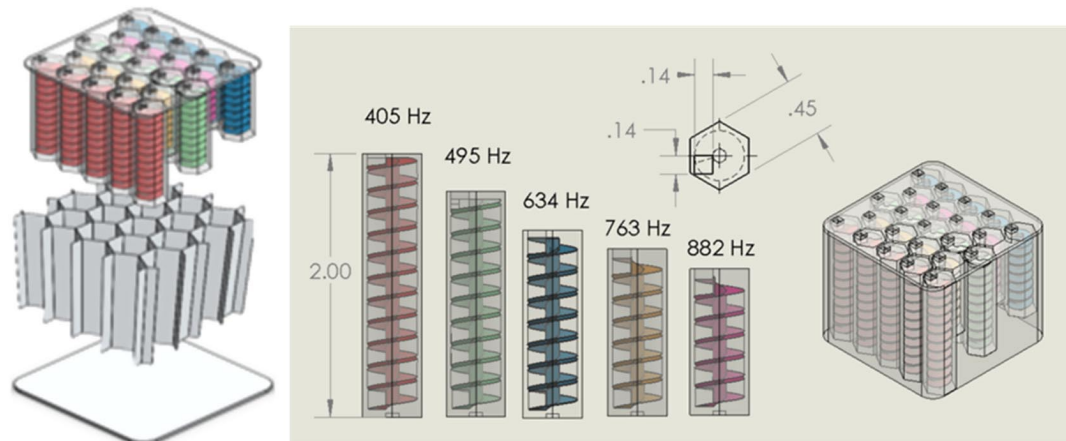


Fig. 10 CAD drawing for the broadband spiral insert liner design. All dimensions shown in inches.

Table 2 Summary of test cases considered in this study

Case No.	Description	Number of resonators (and type)	Liner thickness, mm (in.)	Open area, %	Facesheet
1	Straight cavity baseline	5 × 5 (straight)	44.5 (1.75)	25	No
2	Helmholtz baseline	10 (straight)	50.8 (2)	0.49	Yes
3	U-liner baseline	5 × 5 (straight)	44.5 (1.75)	13.1	No
4	W-liner baseline	5 × 5 (straight)	44.5 (1.75)	11.4	No
5	U-liner	5 × 5 (180° folded)	44.5 (1.75)	13.1	No
6	U-liner with facesheet	5 × 5 (180° folded)	44.5 (1.75)	1.1	Yes
7	W-liner	5 × 5 (3 × 180° folded)	44.5 (1.75)	11.4	No
8	W-liner with facesheet	5 × 5 (3 × 180° folded)	44.5 (1.75)	1.1	Yes
9	Circle spiral liner	5 × 5 (identical spiral)	42.9 (1.69)	4.7	No
10	Square spiral liner	5 × 5 (identical spiral)	42.9 (1.69)	4.7	No
11	Broadband spiral insert liner	23 (multiple spiral)	50.8 (2)	7.2	No
12	Optimized 3D folded liner (OSU)	6 (multiple 180° folded)	38.1 (1.5)	0.47	Yes
13	Optimized 3D folded liner (NASA)	6 (multiple 180° folded)	38.1 (1.5)	0.73	Yes

vicinity of 2000 Hz. The optimization process is described in detail in the modeling section. Based on experimental trials, a wall thickness of at least 1.5 mm (0.0591 in.) was used in this design to ensure the acoustic rigidity for the cavities. It was then possible to accommodate a 6 × 6 cavity array for this case as opposed to 5 × 5 arrays for the previous cases. Two versions of this design were constructed in order to test it in the OSU and NASA tubes. The main difference between the two is that the active area of the liner (the surface area of the face of the liner that the sound wave will encounter) is 50.8 mm × 50.8 mm (2 in. × 2 in.) in the NASA tube as opposed to the 63.5 mm × 63.5 mm (2.5 in. × 2.5 in.) area in the OSU tube. Thus, the optimized 3D folded cavity design was constrained to fit within an effective area of 50.8 mm × 50.8 mm (2 in. × 2 in.) as shown in Fig. 11 in order to be compatible with both tubes. This decrease in active area from the previous folded cavity designs, which were tested only in the OSU tube, is yet another constraint influencing the final number and size of cavities featured in this optimized design. Further, in order to test a particular test article in both the tubes, a simple adaptor sleeve was 3D printed and used to make it fit in the OSU tube. A distinguishing feature of this design is that the total cavity lengths, including partial lengths, are optimized so that the spacing between absorption peaks can be tuned to enhance the design's LFP. Instead of each cluster having an integer number of cells (full depth sections), some cavity lengths have partial cells; however, the packaging of these clusters is done in a manner such that there is no dead volume in the design. Enforcing all these constraints, the optimum cavity lengths were obtained using the 3D folded cavity packing optimization process shown in Fig. 3. Retaining a consistent facesheet thickness of 1.59 mm (0.0626 in.) as before, the optimum hole diameter was obtained from the optimization process to be 2 mm

(0.0787 in.), and this facesheet was printed and tested with this design. Design details and dimensions for this case are shown in Fig. 11. The details of the various test articles considered in this study are summarized in Table 2.

IV. Discussion of Results

Measured and predicted absorption spectra for all of the test cases are presented and discussed in detail in this section. For each design, the experimental spectrum is compared with the ZKTL spectrum for different configurations.

A. Baseline Liners

Figure 12a provides a comparison of experimental absorption spectrum with the ZKTL prediction for the straight cavity baseline case. Good agreement is seen in terms of the location and extent of the peak absorption bandwidth. The peak absorption frequency range for the conventional liner with straight cavities is limited by the liner volume that is available. The peak absorption frequency for the straight cavity baseline liner can be directly estimated as

$$f = \frac{c}{4h_1} \quad (22)$$

where c is the speed of sound and h_1 is the straight cavity's height (or length). Using $c = 340$ m/s (760 mph) and $h_1 = 0.043$ m (1.69 in.) yields a peak frequency of 1977 Hz, which is very close to the experimental and predicted value. Therefore, targeting lower frequencies using straight cavities requires the use of thicker liners, which is not practical. The slight difference in the experimental

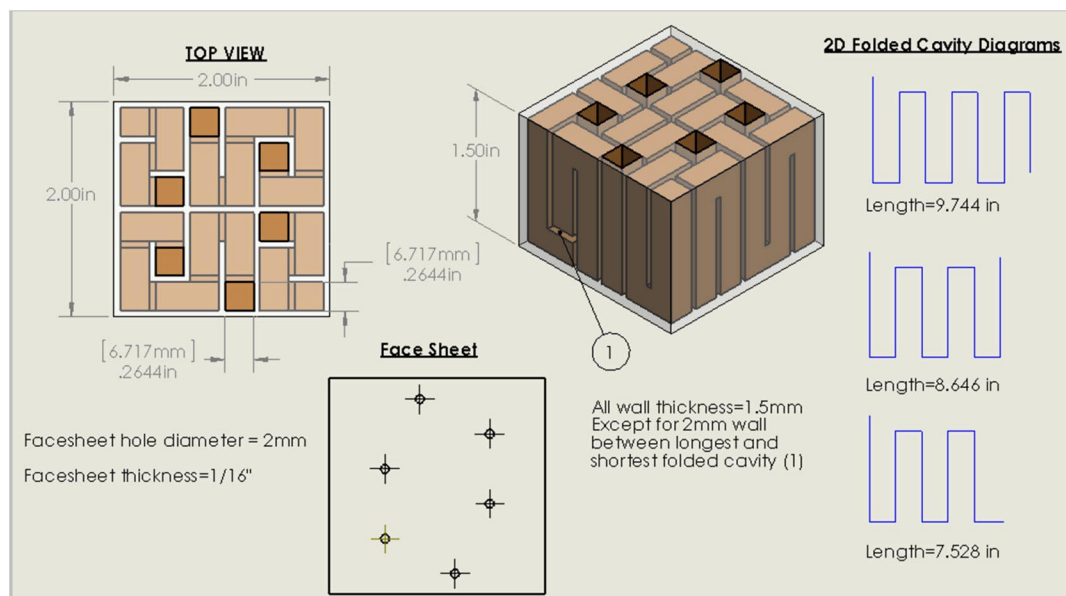


Fig. 11 CAD drawing for the optimized 3D folded cavity liner design. Units for dimensions are specified.

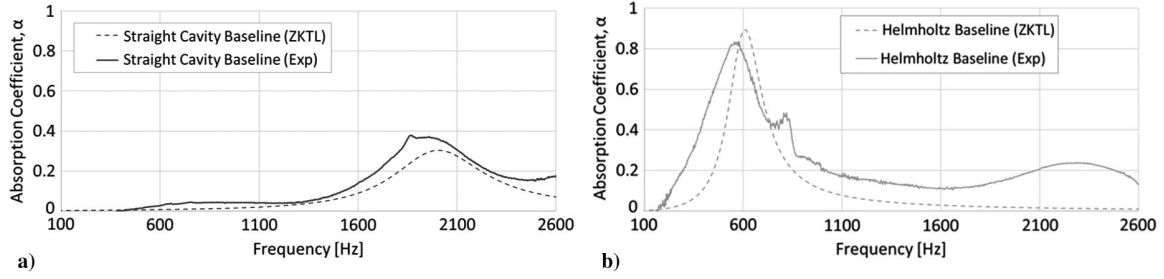


Fig. 12 Predicted and measured absorption spectra from the OSU tube for a) the straight cavity baseline and b) the Helmholtz baseline liners.

spectrum from the simulated spectrum is potentially due to imperfect source and acoustic sealing conditions in the experiments.

Figure 12b shows the comparison of absorption spectra for the Helmholtz baseline case. Again, the absorption peak and bandwidth is predicted with good accuracy. Theoretically, the Helmholtz resonance frequency for the liner cavities at which the peak absorption is expected to occur can be calculated using the relationship

$$f = \frac{c}{2\pi} \sqrt{\frac{S}{Vh_2'}} \quad (23)$$

where S is the cross-sectional area of the liner cavity's neck, V is the volume of the liner cavity, and h_2' is the effective height (or length) of the liner cavity's neck [37]. This effective height for the neck is given by

$$h_2' = h_2 + 0.48(S^2) \left(1 - 1.25 \frac{d_{nc}}{d_{cv}} \right) \quad (24)$$

where h_2 is the height (or length) of the liner cavity's neck, d_{cv} is the cavity diameter, and d is the diameter of the liner cavity's neck. Using the facesheet and core dimensions for the Helmholtz baseline design, Eqs. (23) and (24) give its resonance frequency as about 673 Hz, which is quite close to the location of the peak obtained. The addition of the low-porosity facesheet significantly improves the peak absorption and shifts it to lower frequencies, although it must be noted that the core thickness for the Helmholtz baseline (50.8 mm [2 in.]) is 33.3% greater than the straight cavity baseline (38.1 mm [1.5 in.]). Nonuniformities in the fabricated test article and imperfections in the experimental setup could have contributed to the slight discrepancy in the peak location and extent as well as the occurrence of the minor secondary peak in the experimental results around 830 Hz. The baseline tests demonstrate that thicker liners are required to address lower frequencies, whereas bandwidth of significant absorption is relatively narrow for these cases.

B. Preliminary Folded Cavity Liners

Figure 13a shows that excellent correlation is obtained between experimental and ZKTL simulation results for all three configurations of the U-liner. The U-liner nearly doubles the cavity length compared with its straight cavity baseline without an increase in liner volume or thickness. With the addition of the facesheet peak absorption at around 700 Hz is obtained. A harmonic of the fundamental resonance for the U-liner with the facesheet is also observed within the frequency range of testing. The slight "double peak" predicted in the no-facesheet case stems from modeling the slightly reduced cross-sectional area for the rotated cavity in each row for the U-liner. This feature can also be seen in the experimental spectrum.

Similarly, for the W-liner cases, excellent agreement between the ZKTL predictions and the experimental results is shown in Fig. 13b. For the W-liner, the total cavity length is nearly four times its straight cavity baseline. Both the W-liner and the W-liner with facesheet configurations show a significant shift in peak absorption toward lower frequencies from the straight cavity baseline configuration. The first peak for the W-liner configuration with facesheet is located at ~430 Hz, which falls within the targeted low-frequency range below 500 Hz. The measured LFP for the W-liner is more than 24. Thus, sub-500 Hz absorption peaks are obtainable with the W-liner without any increase in liner thickness over the straight cavity baseline. With reduction of the fundamental frequency in this design, the occurrence of higher harmonics corresponding to odd multiples of the total cavity length are apparent within the testing frequency range and predicted accurately. Optimizing combinations of multiple folded cavities of different total cavity lengths could further widen the bandwidth of absorption above the threshold value at these low frequencies, although this would come at the expense of some reduction in peak absorption within this bandwidth due to there being lower number of resonators at each frequency. The results for the preliminary versions of the folded cavity liners illustrate the potential to achieve enhanced absorption bandwidths at much lower frequencies than conventional liners of the same thickness.

C. Spiral Cavity Liners

With the spiral cavity liner designs, the motivation is to maximize the acoustic path within the available liner thickness in order to achieve peak absorption at lower frequencies than before. The results for the circle spiral cavity liner design are shown in Fig. 14a. This design has a total cavity length greater than that of the W-liner, although to keep the liner volume the same, the cross section of the cavities is smaller. The first peak location is around 370 Hz; however, the measured LFP (5.82) is small due to the peak magnitude and bandwidth of significant absorption ($\alpha > 0.6$) being diminished. The peak locations from the experimental results match very well with the ZKTL simulation. Unlike most of the previous cases, the magnitude of the experimental peak absorption is slightly less than the predicted value. This is likely due to inaccuracies from 3D printing relatively small holes using an extrusion-type method, resulting in higher scrubbing losses. Occurrence of higher harmonics can also be seen as noted before.

Figure 14b shows absorption spectra for the square spiral cavity liner design. A more compact packing of the acoustical path is attempted in this design. As such, the first absorption peak is expected at a lower frequency than the circle spiral case. This peak is expected around 265 Hz, but occurs at a slightly higher value in the experiments likely due to lack of uniformity between individual cavity lengths in the test article. This slight mismatch in the first

Table 3 Low-frequency performance (LFP) parameters for the optimized 3D folded liner test article for the NASA tube

Parameter	Value		
	Sim	Exp	Unit
Lowest continuous bandwidth with absorption $\alpha > 0.6$, β	104	102.8	Hz
Maximum peak value of α within the bandwidth β , M	0.894	0.9	—
Lower bound of the bandwidth β , ζ	290	287.5	Hz
Low-frequency performance metric, LFP	32.1	32.2	—

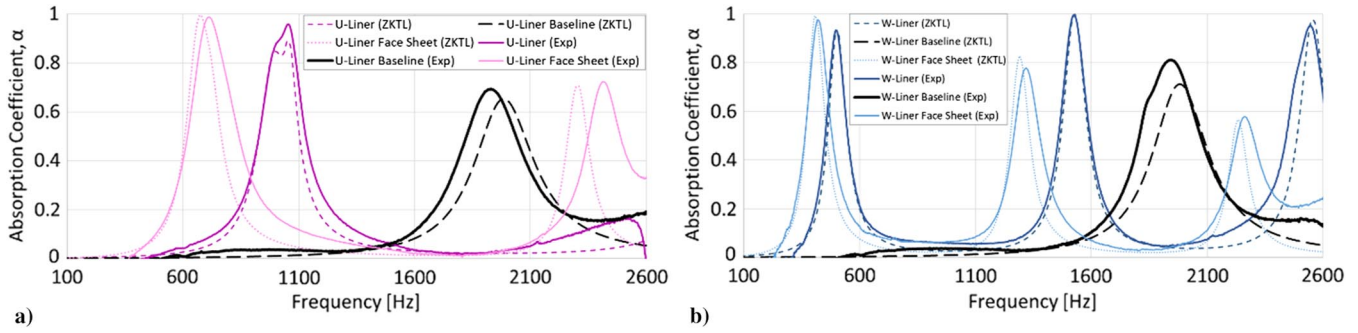


Fig. 13 Predicted and measured absorption spectra from the OSU tube for a) the U-liner and b) the W-liner.

absorption peak frequency between experiment and prediction is naturally magnified for the peaks associated with the higher harmonics as can be seen from the spectra. Again, scrubbing losses within relatively narrow cavities having nonuniform cavity wall surfaces due to the extrusion printing process result in a lower measured value for the peak absorption. Although the first peak location is within the targeted low-frequency range, its magnitude (for the chosen facesheet properties) never exceeds the significant absorption threshold of 0.6 required for calculation of the LFP.

The spectra for the broadband spiral insert liner in Fig. 14c shows that the peak locations from the experimental results match quite well with the peak locations predicted by the ZKTL simulation results. However, the magnitude of absorption is much less than the predicted value for reasons surmised from the results for the earlier spiral designs. Because of the lack of a bandwidth where absorption is significant, measured LFP is not applicable in this case. Nonetheless, the possibility of coalescing absorption bandwidths associated with individual cavity lengths by spacing them appropriately within the targeted frequency range is demonstrated. With better accuracy in the fabrication processes used to make the spiral cavity liners, deterioration in peak absorption could be avoided so as to make this approach more feasible for low-frequency applications.

D. Optimized 3D Folded Cavity Liner

The comparison of the measured and predicted absorption spectra for the optimized 3D folded cavity liner design is shown in Fig. 15a. Test cases from both the OSU and NASA tubes are included. It should be noted that the active area of the test article for the OSU tube is 63.5 mm × 63.5 mm (2.5 in. × 2.5 in.), whereas for the NASA tube, it is 50.8 mm × 50.8 mm (2 in. × 2 in.). Therefore, in both experiments and simulations, it is seen that due to the reduction in

the active area of the test liner from the OSU tube to the NASA tube, although the cavity volumes remain unaltered, the magnitude of the absorption spectra increases commensurately across the entire spectrum of interest without any shift in frequencies associated with specific peaks. It must also be noted that the simulations did not include any nonlinear effects due to differences in sound pressure level. The LFP parameters for the optimized 3D folded cavity liner for the NASA tube are shown in Table 3.

Although good correlation is observed between the predicted and measured absorption spectra, the resonance peaks seen in the predictions due to the optimal spectral spacing of the 3D folded cavities of different lengths in each of the three pairs of clusters are indistinguishable in the experiments. It was noted that the 3D printed facesheet's hole diameter was closer to 1.8 mm (0.0709 in.) instead of the design value of 2 mm (0.0787 in.). Because the facesheet was attached to the core using a layer of double-sided tape, the thickness of the facesheet was estimated to increase to ~2 mm (~0.0787 in.) from 1.59 mm (0.0626 in.). These dimensional changes were accounted for in the ZKTL simulation to more accurately predict the performance. Measured absorption from the OSU tube at an SPL of 110 dB matches its simulation fairly well, as does the NASA test at an SPL of 120 dB. As expected, the magnitude of absorption in the OSU tests was lower than that in the NASA tests although without any shift in frequency, due to the difference in active areas between the tubes. The LFP for the NASA test case is 32.1. Because of the nonlinearity of the perforate facesheet, significant nonlinearity is noticed in the NASA test when the SPL was increased to 140 dB (e.g., the resistance is increased). While the peak absorption was diminished by about 30%, the increase in SPL also results in higher absorption outside of the resonance-driven bandwidth targeted by the folded configuration.

To gauge the sensitivity of the predictions to facesheet parameters, simulations with perturbed inputs for the facesheet hole diameter for

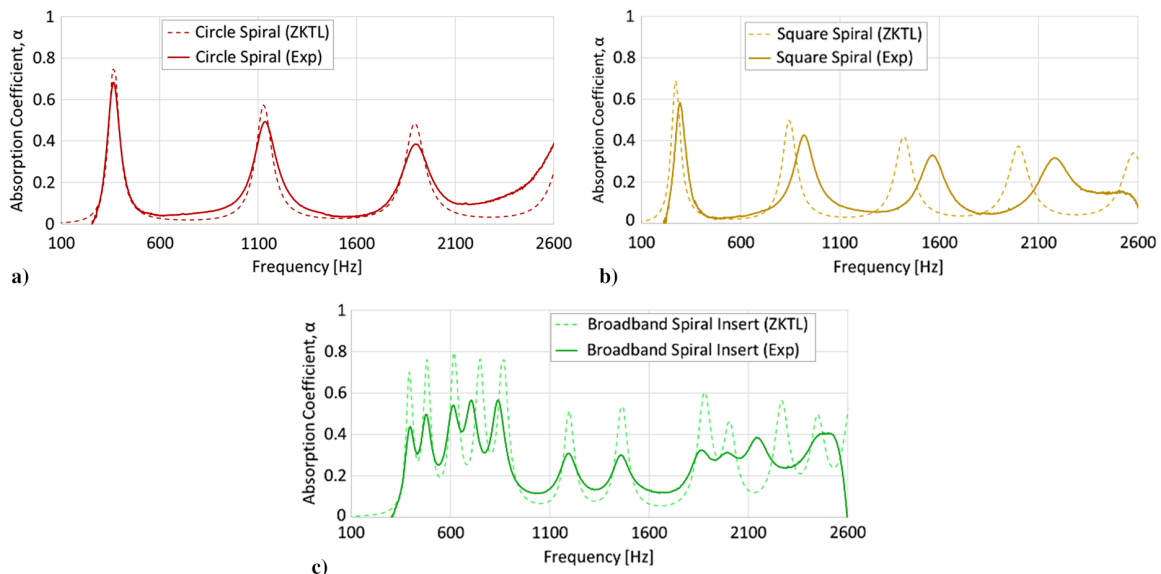


Fig. 14 Predicted and measured absorption spectra from the OSU tube for a) the circle spiral, b) the square spiral, and c) broadband spiral insert liners.

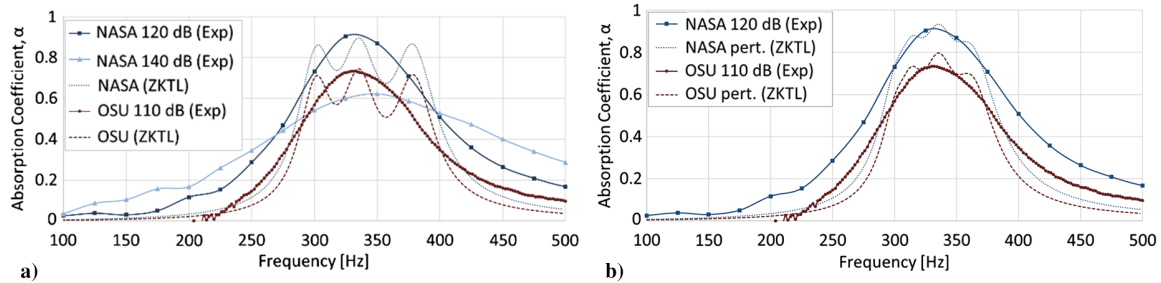


Fig. 15 Comparison of measured absorption spectra for the optimized 3D folded cavity liner with predictions using a) designed and b) perturbed values of facesheet porosity.

each cluster were also done. Owing to the nonuniformity arising from the 3D printing process for the facesheet whose key dimensions (in this case) are comparable to the printer's resolution, it is reasonable to explore their effects in the simulations. Based on observed variations in the 3D printed facesheets, the facesheet hole diameters for each cluster in the 6-cluster configuration were perturbed by amounts within $\pm 20\%$. Therefore, instead of having 3 pairs of identical clusters, the facesheet parameter inputs were revised in the simulations altering their resonant behavior. As can be observed from the simulated absorption spectrum in Fig. 15b, this does change the LFP, which is now 26.7. Comparison with experiments shown in Fig. 15b, however, reveals that the predicted absorption spectrum does indeed become more smeared akin to the experiments when the type of variation observed in the facesheet hole diameter arising from the 3D printing process is taken into account. Peak absorption within the targeted bandwidth shows close agreement with the tuned resonances in the simulations. However, it is conjectured that dissipative contributions to absorption over the entire frequency range render the measured absorption higher than the prediction outside of the resonance-driven bandwidth. Overall, the ZKTL-based optimization procedure using the LFP metric provides a fairly accurate design methodology to realize 3D folded cavity acoustic liners that could provide a feasible solution to effectively mitigate low-frequency noise.

V. Conclusions

A design and optimization procedure based on Zwikker-Kosten Transmission Line (ZKTL) simulations was developed for acoustic liners with 3D folded core geometries in order to enhance low-frequency ($< \sim 500$ Hz) absorption for a specified liner thickness. The relative performance of various designs was compared using a quantitative metric extracted from the absorption spectrum termed the *low-frequency performance* (LFP) metric. Tests were conducted using 3D printed test articles using normal incidence impedance tubes. Maximizing the acoustical path length of folded cavities while retaining cavity cross sections large enough to preclude scrubbing losses results in significant absorption bandwidths at low frequencies without requiring an increase in overall liner thickness. By efficiently packing 3D folded cavities into a prescribed liner volume, exceptional broadband absorption is demonstrated to be achievable at frequencies below 500 Hz with a liner 38.1 mm (1.5 in.) deep. The combination of the LFP metric and the ZKTL-based optimization procedure yields noise absorption solutions tailorable for specific low-frequency bandwidths. Optimizing the relative lengths of the folded cavities to tune the peak locations within the absorption coefficient spectrum can enhance the bandwidth of absorption that the liner exhibits. More than 100 Hz of continuous bandwidth with absorption coefficient greater than 0.6 is shown to be possible in the 250–400 Hz range with a liner 38.1 mm (1.5 in.) deep. It was demonstrated that the ZKTL-based optimization procedure using the LFP metric as a performance metric provides a fairly accurate design methodology to realize 3D folded cavity acoustic liners that could provide lightweight, compact, and structurally integrated solutions to effectively mitigate low-frequency noise. With additive and hybrid manufacturing techniques attaining critical commercial maturity in recent times, 3D folded cavity liners could provide a

promising practical solution to mitigate low-frequency airborne noise, especially in aerospace applications.

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